

Colonic Bacterial Composition in Parkinson's Disease

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ABSTRACT: **Introduction:** We showed that Parkinson's disease (PD) patients have alpha-synuclein (α -Syn) aggregation in their colon with evidence of colonic inflammation. If PD patients have altered colonic microbiota, dysbiosis might be the mechanism of neuroinflammation that leads to α -Syn misfolding and PD pathology.

Methods: Sixty-six sigmoid mucosal biopsies and 65 fecal samples were collected from 38 PD patients and 34 healthy controls. Mucosal-associated and feces microbiota compositions were characterized using high-throughput ribosomal RNA gene amplicon sequencing. Data were correlated with clinical measures of PD, and a predictive assessment of microbial community functional potential was used to identify microbial functions.

Results: The mucosal and fecal microbial community of PD patients was significantly different than control subjects, with the fecal samples showing more marked differences than the sigmoid mucosa. At the taxonomic level of genus, putative, "anti-inflammatory" butyrate-producing bacteria from the genera *Blautia*, *Coprococcus*, and *Roseburia* were significantly more abundant in

feces of controls than PD patients. Bacteria from the genus *Faecalibacterium* were significantly more abundant in the mucosa of controls than PD. Putative, "proinflammatory" Proteobacteria of the genus *Ralstonia* were significantly more abundant in mucosa of PD than controls. Predictive metagenomics indicated that a large number of genes involved in metabolism were significantly lower in the PD fecal microbiome, whereas genes involved in lipopolysaccharide biosynthesis and type III bacterial secretion systems were significantly higher in PD patients.

Conclusion: This report provides evidence that proinflammatory dysbiosis is present in PD patients and could trigger inflammation-induced misfolding of α -Syn and development of PD pathology. © 2015 International Parkinson and Movement Disorder Society

Key Words: microbiota; dysbiosis; α -synuclein; colonic mucosa and feces; putative butyrate producing short-chain fatty acids

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Parkinson's disease (PD) is a neurodegenerative disease with no cure. The diagnosis of PD is based on clinical observations, including resting tremor, bradykinesia, rigidity, and postural instability, reflecting loss of dopaminergic neurons in the SN. Symptoms occur only late in the course of the disease, when significant neuronal loss has already occurred and interventions are targeted at alleviating symptoms rather than preventing disease progression. Therefore, there is a need to understand the pathogenesis of PD in its early stages.

The main pathological change associated with PD is the presence of eosinophilic cytoplasmic neuronal

inclusions (Lewy bodies) that stain for alpha-synuclein (α -Syn) protein, as well as α -Syn-positive neuronal processes (Lewy neurites).^{1,2} Evidence suggests that α -Syn deposition in neurons may begin in the neurons of the intestinal submucosa³⁻⁷ and we have demonstrated α -Syn-positive neurons in sigmoid mucosa samples in patients collected 2 to 5 years before the onset of PD symptoms.⁸ We hypothesize that PD is initiated in the intestine either: (1) through inflammation-induced oxidative stress (OS) and the resulting misfolding of α -Syn with the α -Syn pathology spreading to the brain in a prion-like fashion^{9,10} or (2) as a consequence of intestine-derived lipopolysaccharide (LPS) endotoxin that produces local/systemic inflammation and OS leading to disruption of the intestinal and blood-brain barriers and enteric/neuroinflammation and α -Syn pathology.

It is known that aggregation of α -Syn in the brain is a consequence of OS¹¹; thus, it is plausible that OS also promotes α -Syn aggregation in enteric nerves. If so, then a potential source of OS could be translocation of proinflammatory intestinal luminal factors, such as LPS, into the intestinal mucosa and submucosa. Indeed, LPS can cause loss of dopaminergic neurons in cell culture¹² and supranigral infusions of LPS are used to model parkinsonism in rodents.¹³ We have shown that: (1) Patients with early stage/newly diagnosed PD have intestinal hyperpermeability; and (2) disruption of intestinal barrier integrity in PD patients is associated with α -Syn aggregation in the sigmoid mucosa, oxidative injury, and penetration of bacteria into the sigmoid mucosa and submucosa.¹⁴ We hypothesize that intestinal hyperpermeability and inflammation in PD patients is the result of alterations in the intestinal microbiota composition (dysbiosis) based on the following: (1) Colonic bacteria are a primary source of proinflammatory bacterial products; (2) the abundance of these products can be increased as a result of dysbiosis; and (3) dysbiosis can impact intestinal barrier function.

Sigmoid mucosal biopsies and feces from PD and healthy control (HC) subjects show that the microbiota communities in colonic mucosa-associated and fecal samples were markedly different, and the microbial composition in PD and HC subjects were significantly different.

Patients and Methods

Subjects

Healthy control (N = 34) volunteers were recruited by research advertisements at Rush University Medical Center (RUMC; Chicago, IL), whereas PD subjects (N = 38; 12 untreated naïve/26 treated) were recruited from the movement disorders clinic at RUMC. All subjects consented to use of their samples for research.

Unprepped sigmoidoscopy samples were collected at the gastroenterology clinic at RUMC and fecal samples were collected at home. Please refer to Supporting Information (Methods) for further details on sample procurement.

PD was diagnosed according to the UK Brain Bank Criteria.¹⁵ Exclusion criteria for PD subjects: (1) atypical or secondary Parkinsonism, (2) the use of probiotics or antibiotics within three months prior to sample collection, (3) primary gastrointestinal pathology, (4) unstable medical, neurological, or psychiatric illness, (5) low platelet count (<80 k), uncorrectable prolonged prothrombin time (PT; >15 seconds), or history of bleeding that precludes biopsies.

Inclusion criteria for HC subjects was as follows: (1) normal physical exam and blood work; (2) no digestive complaints, symptoms, or disease; (3) no neurodegenerative disease; and (4) no pro- or antibiotic use for at least 3 months before sample collection. HC exclusion criteria were the same as PD exclusion criteria.²⁻⁵ Five HC subjects were on medication for hyperlipidemia (Oretic, Lipitor, and Zextia), hypertension (Diovan, Diltiazem, and Metoprolol), or anxiety/depression (Effexor, Remeron, and Lexapro).

A subset of subjects completed a modified Harvard food frequency questionnaire (FFQ).^{16,17} Univariate statistics and *t* tests (SPSS, IBM Version 22; SPSS, Inc., Chicago, IL) were used to assess macronutrient intake.¹⁸ Constipation was assessed using a standard visual analog scale from 0 to 10, with 0 indicating no symptoms and 10 indicating the most severe constipation. PD symptoms were assessed using the UPDRS and the modified H & Y scale (HYStage).^{19,20}

Interrogation of Intestinal Bacteria

High-throughput amplicon sequencing of the V4 variable region of the microbial 16S ribosomal RNA gene was performed on genomic DNA extracted from colonic mucosa and feces. Each sample sequence set was subsampled (rarefied) to 8,700 sequences.²¹ Differences in abundance were detected using a Kruskal-Wallis' test generating a Benjamini-Hochberg false discovery rate (FDR)-corrected *P* value. In silico community functional predictions were performed using PICRUSt (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States)²² and significant differences in Kyoto Encyclopedia of Genes and Genomes (KEGG) ortholog (KO) abundances between groups were identified (FDR-corrected *P* value).²³ Putative "proinflammatory" and "anti-inflammatory" bacteria taxa were classified based on previous reports (and the references therein).²⁴⁻²⁸ Short-chain fatty acids (SCFAs), especially butyrate-producing bacteria at the intestinal level, play a major regulatory role in the intestinal ecosystem to maintain gastrointestinal function and body health.²⁹⁻³² Please refer to the

Supporting Information (Methods) section for more information about the interrogation of intestinal bacteria and the mapping data for demultiplexing (Table S1).

Results

Subjects

Subject characteristics are shown in Tables S2 and S3. Sex and body mass index (BMI) were indistinguishable between the PD and HC groups (sex [male/female]: PD = 24/14, HC = 18/16, $P = 0.39$; BMI: PD = 26.0 ± 3.2 , HC = 27.6 ± 6.0 , $P = 0.16$). Mean age of PD subjects (61.6 ± 9.4) was greater than that of HCs (45.1 ± 14.4 ; $P < 0.00$). The HC group contained more African-American subjects than the PD group (PD = 35/1/2; HC = 18/11/5 Caucasian/African American/other, respectively; $P < 0.00$); however, microbial composition (feces or mucosa) was not significantly different between HC Caucasian and African Americans (feces: ANOSIM Global R, $P = -0.01$, $P = 0.53$; mucosa: Global R = -0.03 , $P = 0.59$). Five HC subjects were taking medication at the time of sample collection. Removal of the HC subjects on medication did not alter the between group differences (Table S4); thus, these samples were included for all data analysis. PD patients had average disease duration of 6.4 ± 4.7 years, UPDRS of 29.7 ± 11.0 , HYSage of 2.1 ± 0.4 , constipation duration of 15.8 ± 19 years, and constipation severity of 2.8 ± 3.4 . Diet data were collected for 24 PD and 15 HC subjects; there were no significant differences in macronutrients, dietary fiber, or total calorie intake between groups (Table S5). Significant between group differences were observed in microbial composition untreated naïve PD and treated PD (feces: ANOSIM Global R = 0.20, $P = 0.01$; mucosa: Global R = 0.31, $P < 0.00$).

Comparison of Mucosa and Fecal Microbiota (PD and HC)

Significant differences in community diversity (α -diversity) were observed between mucosal-associated and fecal microbiota within PD and HC subjects (Table S6). Likewise, microbial composition in feces and mucosa were significantly different at multiple taxonomic levels (Table S4; Fig. S1). Average relative abundance of sequences from the phylum Proteobacteria was significantly higher in mucosal samples than in fecal samples (Table 1), whereas Firmicutes and the ratio of Firmicutes-to-Bacteroidetes was significantly higher in feces than in mucosal samples (Fig. S2; $P = 0.04$). Putative anti-inflammatory butyrate-producing bacteria (genera *Blautia*, *Roseburia*, and *Coprococcus*) were significantly more abundant in fecal, compared to mucosal, samples (Fig. 1; Table 1).

HC Versus PD Subjects: Mucosa-Associated Bacteria

The α -diversity of mucosal-associated bacterial communities was, for the most part, not altered between PD and HC subjects (Table S6); only the family level demonstrated a significant difference in richness (i.e., the number of species per sample). However, the composition of mucosal-associated microbial communities was significantly different between HC and PD subjects (Tables 1 and S4). Bacteria from the family Coprobacillaceae (Firmicutes, class Erysipelotrichi), genera *Dorea* (Firmicutes, class Clostridia), and anti-inflammatory *Faecalibacterium* (Firmicutes, class Clostridia) were significantly more abundant in HC, compared to PD, subjects. Putative proinflammatory bacteria from the family Oxalobacteraceae (Proteobacteria, class Betaproteobacteria) and genus *Ralstonia* (Proteobacteria, class Betaproteobacteria, family Oxalobacteraceae) were significantly more abundant in PD subjects (Table 1). Of the butyrate-producing bacteria, only those from the genus *Faecalibacterium* (Firmicutes, class Clostridia) were significantly decreased in PD mucosa (Fig. 1). There was no significant difference in the ratio of Firmicutes-to-Bacteroidetes between mucosal PD and HC samples (Fig. S2; $P = 0.33$).

HC Versus PD Subjects: Fecal Microbiota

Diversity of fecal bacterial communities was not largely different between PD and HC subjects. Significant differences in α -diversity between PD and HC groups were primarily observed at the phylum level, and a significant difference in richness was observed at the genus level (Table S6). More PD-associated changes were observed in feces, compared to mucosal, samples. The relative abundance of four phyla, five families, and seven genera was significantly different between PD and HC fecal samples (Tables 1 and S4). At the phylum level, Bacteroidetes, Proteobacteria, and Verrucomicrobia were higher in PD samples, whereas Firmicutes were higher in HC samples. The ratio of Firmicutes-to-Bacteroidetes in fecal samples was not significantly different between PD and HC groups, though a trend toward a lower ratio in PD patients was observed (Fig. S2; $P = 0.09$). At the genus level, putative proinflammatory bacteria *Akkermansia* (Verrucomicrobia, class Verrucomicrobiae), *Oscillospira* (Firmicutes, class Clostridia), and *Bacteroides* (Bacteroidetes, class Bacteroidia) were significantly more abundant in PD fecal samples. Putative anti-inflammatory SCFA butyrate-producing bacteria *Blautia*, *Coprococcus*, and *Roseburia* (Firmicutes, class Clostridia) were significantly less abundant in PD versus HC feces (Fig. 1; Table 1).

TABLE 1. Relative abundance of sequences derived from individual taxa

Taxonomic Level	Mucosa vs. Feces			Mucosa: PD vs. HC			Feces: PD vs. HC		
	Mucosa # Seqs	Feces # Seqs	FDR- <i>P</i> Value ^a	HC # Seqs	PD # Seqs	FDR- <i>P</i> Value ^a	HC # Seqs	PD # Seqs	FDR- <i>P</i> Value ^a
Phylum									
Actinobacteria	176	255	0.50	225	137	0.53	336	181	0.10
Bacteroidetes	2,518	2,228	0.14	2,457	2,566	0.66	1,775	2,641	0.03 ^a
Firmicutes	4,296	5,766	0.00 ^a	4,400	4,214	0.60	6,340	5,243	0.03 ^a
Proteobacteria	1,352	163	0.00 ^a	1,249	1,432	0.53	112	209	0.03 ^a
Verrucomicrobia	179	206	0.14	176	180	0.42	83	317	0.03 ^a
Family									
Bifidobacteriaceae	106	250	0.05 ^a	162	62	0.27	331	176	0.20
Bacteroidaceae	1,587	1,532	0.38	1,399	1,735	0.33	1,158	1,873	0.05 ^a
Porphyromonadaceae	298	133	0.00 ^a	207	370	0.43	83	178	0.20
Prevotellaceae	445	143	0.00 ^a	627	302	0.78	142	144	0.88
Rikenellaceae	91	260	0.00 ^a	92	91	0.73	232	286	0.30
Clostridia (c)	77	85	0.05	67	85	0.43	87	82	0.55
Clostridiaceae	574	49	0.00 ^a	367	737	0.27	33	64	0.05 ^a
Lachnospiraceae	1,647	2,168	0.06	1,700	1,606	0.78	2,754	1,634	0.02 ^a
Peptostreptococcaceae	45	214	0.00 ^a	49	42	0.80	291	144	0.40
Ruminococcaceae	1,266	2,192	0.00 ^a	1,424	1,141	0.27	2,050	2,322	0.31
Veillonellaceae	189	171	0.06	201	179	0.78	137	201	0.39
Coriobacteriaceae	39	207	0.00 ^a	38	40	0.27	189	223	0.40
[Coprobacillaceae]	101	153	0.02 ^a	166	50	0.01 ^a	238	76	0.05 ^a
Erysipelotrichaceae	101	122	0.58	145	67	0.33	204	48	0.26
Oxalobacteraceae	203	1	0.00 ^a	38	332	0.00 ^a	1	1	0.83
Enterobacteriaceae	519	66	0.00 ^a	408	606	0.91	30	99	0.40
Pseudomonadaceae	379	1	0.00 ^a	511	275	0.43	2	1	0.55
Verrucomicrobiaceae	180	206	0.13	178	181	0.27	85	317	0.05 ^a
Genus									
Bifidobacterium	102	252	0.02 ^a	154	61	0.48	335	175	0.14
Bacteroides	1,586	1,527	0.38	1,397	1,733	0.48	1,156	1,866	0.05 ^a
Parabacteroides	220	133	0.24	182	251	0.80	82	178	0.14
Prevotella	444	143	0.00 ^a	626	301	0.72	142	144	0.88
Rikenellaceae (f)	91	260	0.00 ^a	92	91	0.66	233	284	0.27
Clostridia (c)	77	85	0.05	66	85	0.52	87	83	0.58
Clostridiaceae (f)	213	36	0.02 ^a	123	283	0.48	28	44	0.09
Lachnospiraceae (f)	428	454	0.26	360	482	0.55	528	388	0.27
[Ruminococcus]	293	312	0.40	298	289	0.77	379	251	0.27
Blautia (B)	459	790	0.01 ^a	490	434	0.60	1,032	570	0.03 ^a
Coprococcus (B)	86	169	0.00 ^a	97	78	0.60	241	103	0.03 ^a
Dorea	134	60	0.35	178	100	0.01 ^a	88	35	0.01 ^a
Roseburia (B)	173	250	0.02 ^a	195	155	0.52	341	167	0.03 ^a
Peptostreptococcaceae (f)	37	213	0.00 ^a	40	35	0.74	290	143	0.33
Ruminococcaceae (f)	330	715	0.00 ^a	375	295	0.58	665	760	0.43
Faecalibacterium (B)	389	572	0.11	543	269	0.02 ^a	645	504	0.30
Oscillospira	130	249	0.00 ^a	131	130	0.55	189	304	0.03 ^a
Ruminococcus	389	608	0.00 ^a	355	415	0.75	506	702	0.27
[Eubacterium] (B)	92	111	0.94	134	60	0.48	185	43	0.14
Ralstonia	183	1	0.00 ^a	26	307	0.00 ^a	1	0	0.61
Escherichia	443	52	0.00 ^a	302	554	0.60	23	78	0.36
Pseudomonadaceae (f)	371	1	0.00 ^a	502	269	0.55	2	0	0.30
Akkermansia	179	206	0.17	177	180	0.48	83	318	0.03 ^a

() = classification to higher taxonomic level ((c) = class; (f) = family). (B) = putative SCFA butyrate-producing bacteria. # seqs = average number of sequences in defined group.

^aNonparametric Kruskal-Wallis' one-way analysis of variance generating Benjamini-Hochberg false discovery rate (FDR-*P*)-corrected *P* value: *P* < 0.05 (also highlighted in gray).

Clinical Correlations With the PD Microbiota

We correlated microbial composition with PD duration, disease severity (UPDRS and HYStage), constipation, BMI, and age. PD duration correlated with the greatest number of taxa, with Bacteroidetes and Pro-

teobacteria being positively correlated and Firmicutes negatively correlated (Fig. 2; Table S7). We observed a significant negative correlation between PD duration and the family Lachnospiraceae (feces: Pearson's *R* = -0.51; *P* = 0.02), which is comprised of several anti-

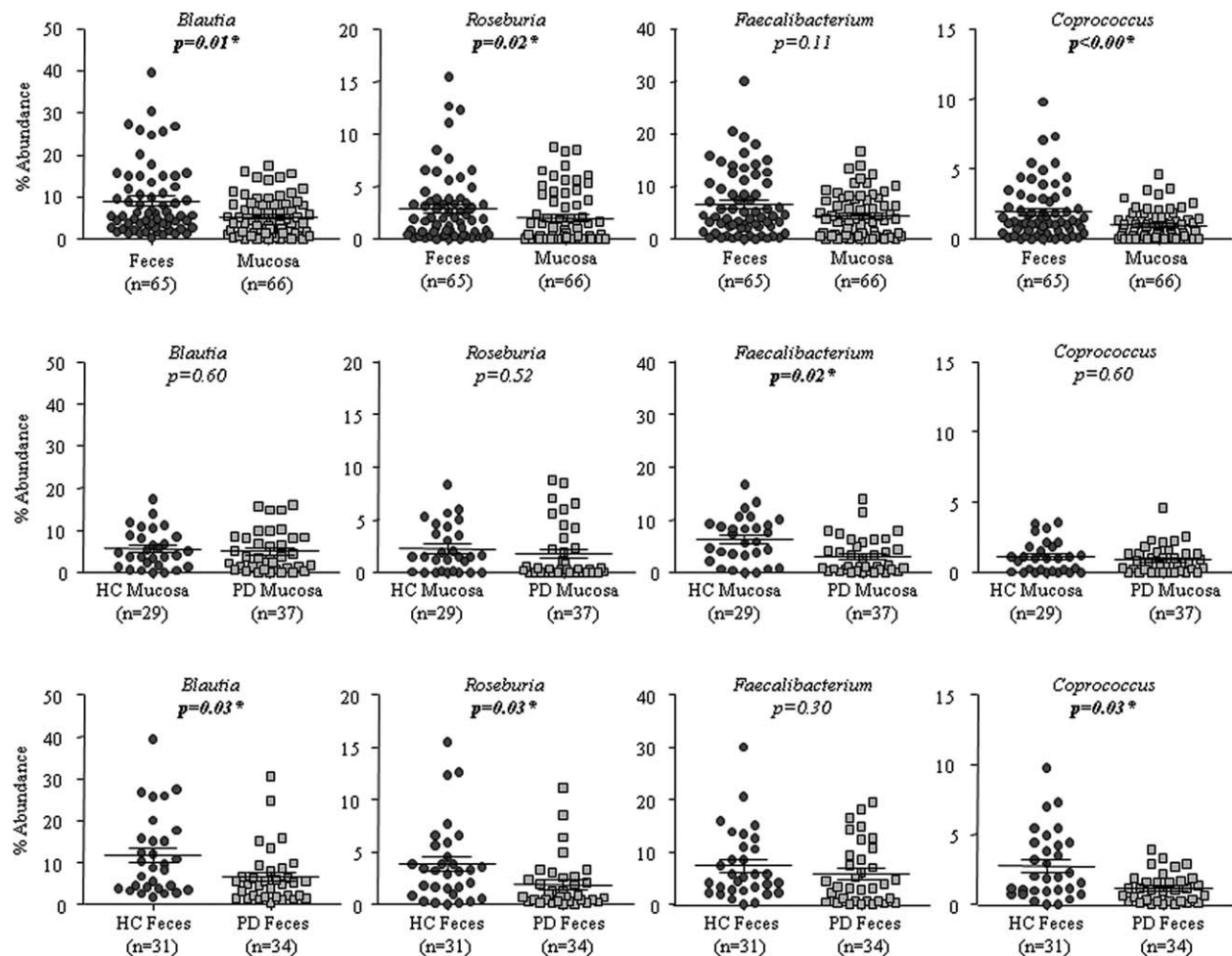


FIG. 1. Percent abundance of putative SCFA butyrate-producing bacteria. Plots represent the putative SCFA butyrate-producing bacteria that were significantly increased in fecal and mucosal samples in comparison to all combined samples and to PD and HC subjects. * $P < 0.05$.

inflammatory butyrate-producing bacteria, including *Blautia*, which also showed a trend for a significant negative correlation (feces: Pearson's $R = -0.45$; $P = 0.09$; Table S7). When untreated naïve PD samples were omitted from the analysis, a significant correlation was still maintained between various bacterial taxa and PD duration (data not shown). No significant correlations were observed between PD disease severity and bacterial taxa. Of the 296 correlations between taxa, age, and BMI (feces and mucosa), we found only nine significant correlations: age correlated with two taxa in PD and none in HC subjects and BMI correlated with five taxa in PD and two in HC subjects (data not shown).

Predictive Functional (PICRUSt) Analysis

Using a predictive assessment of the microbial community functional potential (PICRUSt) we identified 1,248 KOs that had significantly different abundances between PD and HC fecal samples (FDR, $P < 0.05$; Table S8). Most reference pathways had more genes in HC subjects, relative to PD patients, particularly

metabolic and biosynthetic pathways (Fig. 3A). Conversely, pathways including bacterial secretion, LPS biosynthesis, and ubiquinone biosynthesis were more abundant in PD, relative to HC feces (Fig. 3A; Table S9). Fourteen KOs within the LPS biosynthesis pathway and eight within the ubiquinone and other terpenoid-quinone biosynthesis pathway were significantly more abundant in PD feces than in HC feces. Moreover, genes from type III secretion systems, typically associated with pathogenic bacteria,³³ were more abundant in the microbiome of PD feces, whereas genes from other secretion systems were slightly (but significantly) more abundant in HC fecal microbiomes (Fig. 3B; Table S9).

Discussion

Significant differences were observed at all taxonomic levels between mucosal-associated and fecal microbiota communities, a difference not unusual considering the environmental differences between the mucosa and the lumen.³⁴ Our findings also demonstrate that colonic

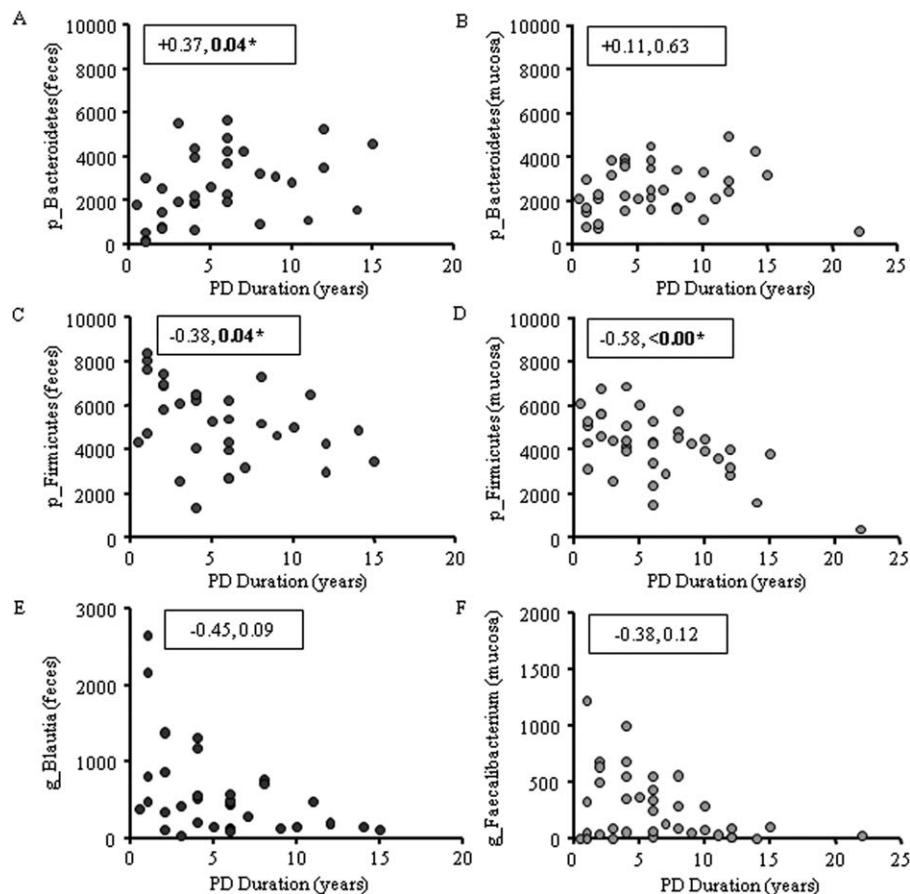


FIG. 2. PD duration correlated to *Firmicutes* to *Bacteroidetes* in mucosal and fecal samples. *Bacteroidetes* (A and B) were positively and *Firmicutes* (C and D) negatively correlated with PD duration. SCFA producers *Blautia* (E) and *Faecalibacterium* (F) both negatively correlated with PD duration. * $P < 0.05$.

mucosa and fecal microbiota collected from PD subjects are different from HC subjects. The dysbiotic changes observed in PD subjects would be expected to impact inflammation given that dysbiosis can disrupt intestinal barrier function and initiate immune activation and systemic inflammation.³⁵⁻³⁸ Exposure of immune cells to the colonic microbiota educates mucosal immune cells that can then travel to the proximal gastrointestinal (GI) tract (and other organs) to modulate immune and inflammatory responses.^{35,39-41} PD pathogenesis may be caused/exacerbated by (1) dysbiotic microbiota-induced inflammatory responses could promote α -Syn pathology in the intestine and brain or (2) rostral to caudal cell-to-cell transfer of α -Syn pathology caused by increased OS burden (i.e., increase in proinflammatory bacteria).^{42,43}

Environmental factors associated with PD (age, BMI, medication use, and PD duration) might explain differences between our PD and HC subjects. We found only seven correlations between BMI and bacterial taxa in the colonic mucosa and feces and only two significant correlations between age and bacteria taxa. Thus, although there was a significant difference in age between PD and HC subjects in our study, age

seems to have little impact on bacterial composition in feces or mucosa in our subject cohort. Nonetheless, we acknowledge the potential limitation of age in our study.⁴⁴⁻⁵⁰ Similarly, there were differences in race between our PD and HC cohorts and whereas race can impact microbiota,^{46,49,50} we did not observe differences between Caucasian and African-American HC subjects, suggesting no effect of race in our cohort. Finally, dietary analyses revealed no between-group differences in macronutrients, dietary fiber, and total calorie intake.

Our study included samples from untreated naïve PD subjects as well as treated PD patients. Microbiota composition (feces and mucosa) in these PD cohorts were significantly different, an effect that may be attributed to PD medication use or disease duration. We observed a significant correlation between numerous bacterial taxa and PD duration, and this significance was maintained when untreated naïve samples were omitted from the analysis, suggesting that though medication is likely having an effect, disease duration is still significantly impacting microbial communities in PD patients. These data are not longitudinal; thus, it is not possible to determine whether dysbiosis is a

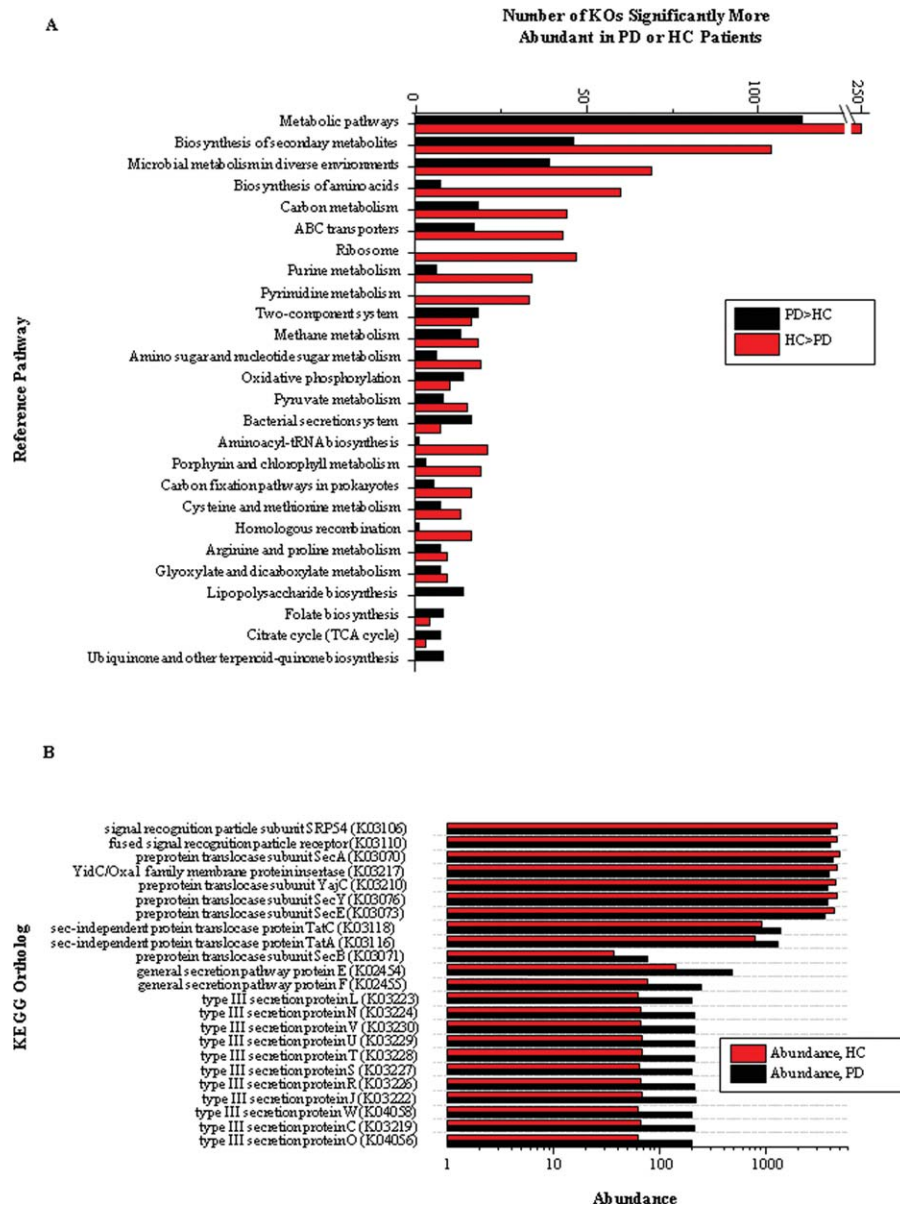


FIG. 3. Functional predictions for fecal microbiome of HC and PD subjects. (A) Significantly different abundant KOs were identified using the software package PICRUSt and classified into pathways as described in the text. Pathways with the greatest number of differently abundant KOs are shown. (B) Significantly different KOs within the bacterial secretion system pathway are shown; KOs associated with type III secretion proteins were consistently more abundant in PD subject microbiomes relative to HC subject microbiomes. TCA, tricarboxylic acid cycle. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

cause or consequence of PD pathogenesis. In either case, we propose that dysbiosis may still play a role in neuronal loss by perpetuating inflammatory cascades and oxidative injury in the brain through an LPS-mediated mechanism.

Intestinal bacteria may play a role in PD intestinal symptoms, such as constipation and bloating.⁵¹⁻⁵⁵ Constipation associated with PD may impact microbiota composition; however, only five significant correlations were observed between constipation score and taxa in the feces and none in the colonic mucosa-associated community, suggesting that, in our subject cohort, constipation had little effect on microbiota.

Our findings are limited to colonic mucosa and fecal bacteria microbiota. PD is associated with numerous GI abnormalities, including small intestinal bacterial overgrowth and *Helicobacter pylori* infection of the stomach. This may be associated with irregularities in levodopa absorption/response, yet no studies have analyzed the microbiota in the small intestine or stomach in PD.⁵⁶⁻⁶⁰ Future studies characterizing microbiota in other parts of the GI tract are needed to determine the influence of the upper GI tract on microbiota in PD patients.

We propose that intestinal dysbiosis and gut leakiness drive neuroinflammation associated with the

pathogenesis of PD. We found that *Blautia*, *Roseburia*, and *Coproccoccus* were less abundant in PD fecal samples than HC. The lower abundance of these SCFA butyrate-producing bacteria could have harmful effects for the PD colon, including intestinal barrier integrity and immune function.^{31,32} The main site of action for SCFA appears to be the colon, but SCFAs are also taken up in the blood and exert effects on other organs, including the brain.⁶¹ SCFA butyrate has anti-inflammatory properties thought to be owing to an epigenetic mechanism^{62,63} or to the effects of SCFA receptors that are anti-inflammatory, anti-microbial, and decrease intestinal barrier leakiness.⁶⁴⁻⁶⁶ Indeed, PD duration negatively correlated with the family Lachnospiraceae, which contains a number of butyrate-producing bacteria. Thus, the low abundance of butyrate-producing bacteria in feces from PD subjects may be one mechanism contributing to intestinal leakiness⁶⁷ and inflammation reported in PD.¹⁴ Metabolomic studies will be required to substantiate the role of SCFA in intestinal inflammation and barrier dysfunction in PD, and metagenomic and -transcriptomic studies will be required to define specific strains of SCFA-producing bacteria, as well as to verify expression of genes in SCFA-production pathways.

Other changes in PD microbiota may also promote inflammation/immune activation and OS in the intestine. Low abundance of Lachnospiraceae in PD feces are similar to changes observed in patients with inflammatory bowel disease,⁶⁸ an intestinal disease with a high degree of intestinal inflammation, and increased relative abundance of *Ralstonia* (Proteobacteria, class Betaproteobacteria)⁶⁹ in mucosal-associated microbiota community in PD could contribute to neuroinflammation and immune activation in PD patients. The observation that type III secretion systems were enriched in PD feces supports that intestinal dysbiosis in PD promotes neuroinflammation given that type III secretion systems are generally involved in pathogenicity and translocation of proteins that could facilitate bacterial product-induced inflammation.^{33,70}

If dysbiosis or mucosal inflammation drives PD pathogenesis, then conditions associated with dysbiosis or mucosal inflammation may be expected to increase the risk of PD development. There are no reports of increased risk of PD in patients with inflammatory bowel disease; nor are there reports of PD symptoms being improved by inflammatory bowel disease treatments. Genetic studies suggest that PD patients are more likely to have variations in genes associated with Crohn's disease, a type of inflammatory bowel disease, as well as variations in the leucine-rich repeat kinase genes (a gene associated with inherited PD) that are also associated with Crohn's disease risk.^{71,72} However, there are drastic differences between the

magnitude of inflammation in inflammatory bowel disease (inflammation is visible on colonoscopy) and PD (typically normal on colonoscopy), and the type of inflammation in these two disorders may be different. Taken together, it is not clear whether patients with inflammatory bowel disease are more likely to develop PD, but there are interesting features that warrant further investigation.

There is one other report demonstrating changes in fecal microbiota composition in PD patients, which showed a decrease in the family Prevotellaceae in fecal samples.⁷³ We did not observe a significant decrease in the abundance of Prevotellaceae in our fecal analyses; however, the trend was the same in our tissue analyses. The average abundance of bacteria from the genus *Prevotella* within PD tissue samples was less than half that observed in HC samples, but this failed to reach significance (0%–45% in HC and 0%–22% in PD). The discrepancy between our study and the previous study could result from differences in methodology (primer set and polymerase chain reaction conditions) or differences in control subjects. In the previous study, the control cohort included subjects with cardiovascular and cerebrovascular disease, including the use of warfarin. History of cardiovascular disease and medication use may cause changes in diet or exercise that may have impacted microbiota composition.^{74,75} For example, patients taking warfarin are advised to not eat many vegetables.⁷⁶ Therefore, it is plausible that the differences in *Prevotella* could be explained by diet.^{49,77,78}

In summary, we demonstrated that both colonic mucosa-associated and fecal microbiota composition are different in PD, compared to HC, subjects and that the bacterial community in PD feces had significantly altered function. Our study provides scientific rationale to design and conduct interventional studies with pre-/probiotics to correct dysbiosis to determine whether changes in microbiota can impact PD pathology and disease course and/or improve GI symptoms. ■

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Supporting Data

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